Sound reduction estimates

Combined baffle box for the 24 x 24 in Big Fan and two 10 in outlets. Separate LD6 roof route for the 6 in Exhaust.

At the reported ~244 Hz dyno peak | The 250 Hz octave is the closest data band. The box screens at **5.4 to 8.5 dB** for the two bypass cases below and **13.3 dB** in the idealized zero-bypass case. Fantech lists **7 dB** for the LD6 alone in that band.

Octave Hz	Rotax accel. dB	Rotax braking dB	703 FRK alpha	Box 0%	Box 10%	Box 25%	LD6 dB
63	70	70	-	-	-	-	-
125	87	78	0.18	7.7	6.0	4.2	3
250	86	73	0.75	13.3	8.5	5.4	7
500	90	73	0.58	12.2	8.1	5.3	20
1000	84	76	0.72	13.1	8.4	5.4	27
2000	79	73	0.62	12.5	8.2	5.3	31
4000	81	71	0.35	10.2	7.3	4.9	16
8000	81	70	-	-	-	-	11

Published Rotax 125 whole-kart pass-bys are spectral context (acceleration at 9 m, braking at 13 m). Site readings are user-reported peaks around 244 Hz during the dyno release transient. The exact band levels and meter settings were not recorded. A dash means data unavailable.

Box calculation: T = A / (A + alpha S), with outlet A = 5.713 ft2 and estimated exposed lining S = 154.90 ft2. Loss = -10 log10[b + (1-b)T], where b is an assumed untreated energy fraction. This diffuse-energy screen has limited precision near 244 Hz, where the wavelength is about 4.6 ft and the cavity depth is about 1.5 ft. LD6 figures apply to its separate branch.

Marshall Day Acoustics, Appendix 5

Owens Corning Type 703 absorption data

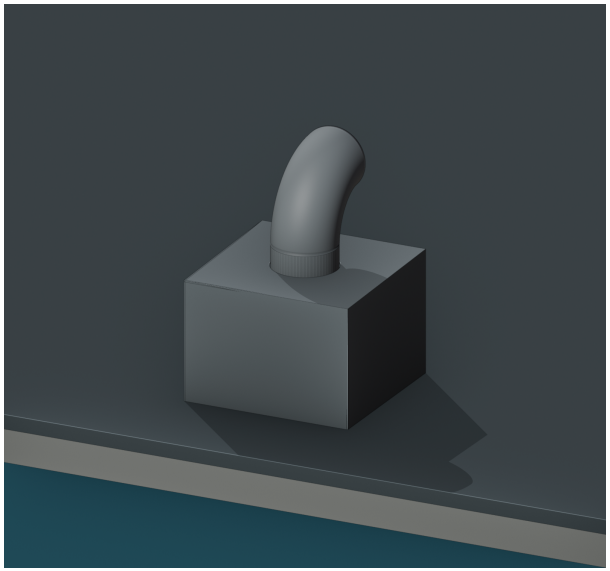
Fantech LD series submittal

Peak readings and roof route

Site readings at the reported ~244 Hz event, with conditional estimates at the same observation positions.

Outlet	iPhone dB	Meter dB	Treatment at 250 Hz	Illustrative same-point meter dB
Big Fan	75	82	Option 4 box	74-77
Inline	85	94	Option 4 box	86-89
Water	70	80	Option 4 box	72-75. Cap present at scan
Exhaust	74	87	Separate LD6	80 from LD6 catalog value

The projected readings subtract the 250 Hz loss from the reported peaks, assuming the treated path dominates at each microphone and conditions repeat. After installation the three box outlets share one discharge. The rows cannot be summed into a road-side level. To estimate the combined level, obtain band levels from each path at the same receiver and sum their sound energies.



6 IN ROOF ASSEMBLY

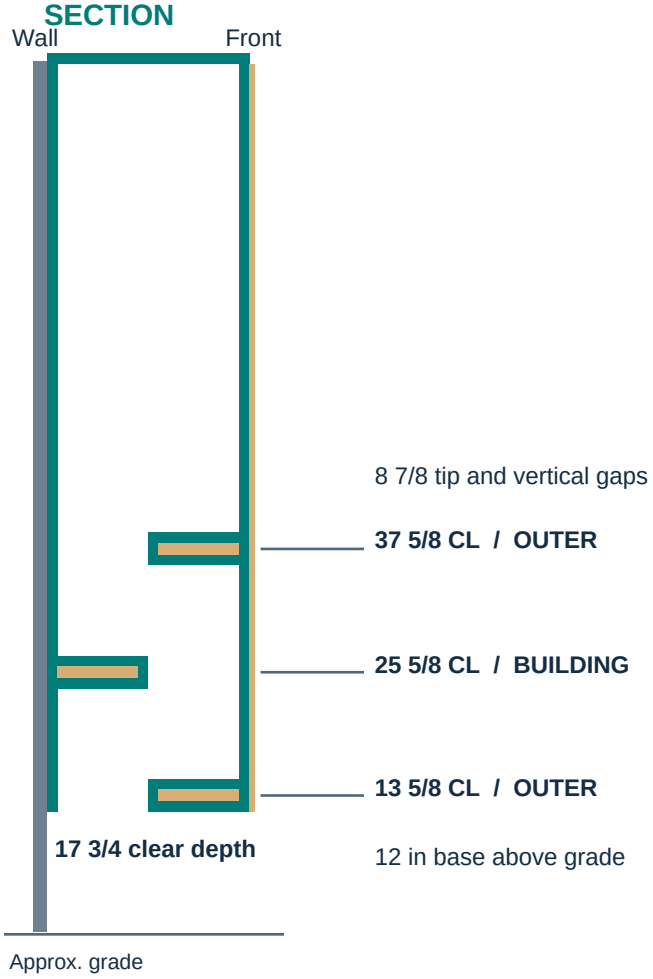
6 in round short-radius 90-degree elbow from wall, vertical LD6, then a 6 in short-radius elbow aimed inward. The 16 in square chase covers the full 10 1/4 in diameter body. The reduced neck and elbow project above the chase. Catalog length shown as 23 5/8 in. Confirm the connection-to-connection dimension, supports, drainage and service conditions.

Roof reflections, outlet direction and pressure drop are additional installation effects. The invalid in-duct microphone reading is excluded.

Box construction and framing

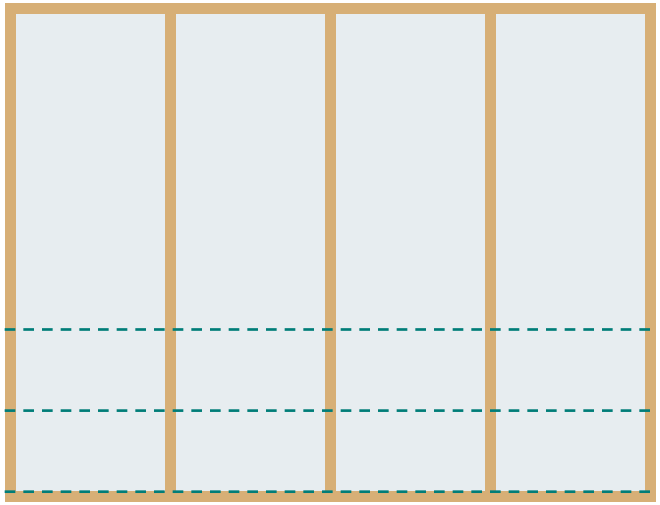
Dimensions in inches, rounded to the nearest 1/8 in. Teal denotes 1 in FRK. Tan denotes the baffle core.

SECTION



FRONT ELEVATION

96 1/2 nominal overall width



73 7/8 shell height

92 7/8 lined clear width

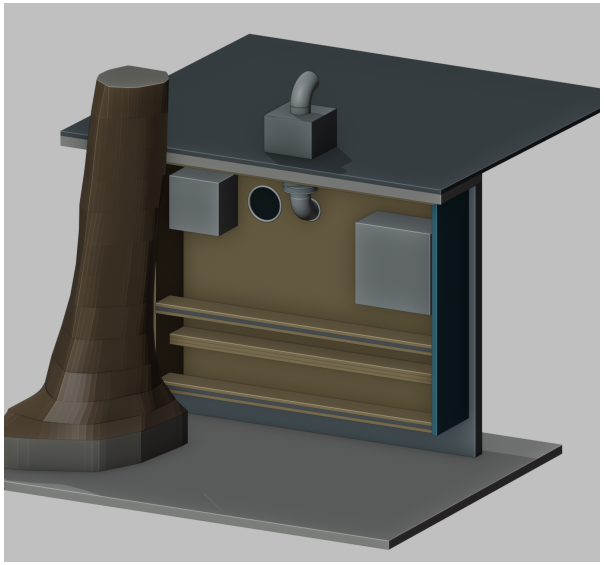
21 wall to modeled outer face

Proposed wall framing: 2 x 3 nominal (1 1/2 x 2 1/2 actual), laid flat with 1 1/2 in outward build. Front layout uses two 96 1/2 in rails and five 68 7/8 in studs between rails (24 in o.c. maximum). Back framing, anchors and baffle supports require site detailing. Keep framing and fasteners clear of the finished flow path and verify the tree-side envelope before selecting the framing layout.

Skin and baffles: 1/2 in exterior plywood. Each baffle uses a 1 1/8 in nominal core envelope (1/2 in plywood plus 5/8 in support allowance), with FRK on both faces and the free tip. Finished thickness is 3 1/8 in and finished projection is 8 7/8 in from its lined mounting wall.

FRK locations and cut sheet

One-inch Owens Corning Type 703 with FRK. Listed sizes are nominal net coverage before scribing and vent cutouts.



Line five enclosure surfaces: building wall, outer front, both ends and ceiling. Keep all vent mouths open. Line both broad faces of each baffle and wrap its free edge with a separate strip. The underside remains open at the discharge slot.

Fit order: establish the finished planes, template hood and duct penetrations, then trim pieces to meet at the corners without stacking layers. Retention, facing orientation and edge treatment must suit the actual airstream and service environment.

Location	Qty	Nominal net panel or strip size	Cut notes
Outer front and building wall	2	92 7/8 x 72 7/8	Gross before field cutouts
Left and right ends	2	17 3/4 x 72 7/8	Scribe to adjoining liner
Ceiling	1	92 7/8 x 17 3/4	Fit around penetrations
Baffle top and bottom faces	6	92 7/8 x 7 7/8	Two faces per baffle
Baffle free tips	3	92 7/8 x 1 1/8	Fits between face layers

Sheet layout: For 24 x 48 in stock, split long pieces at 46 7/16 in. Main walls use three 24 in courses plus a 7/8 in strip from offcuts. Split each end into 48 in and 24 7/8 in lengths. Four half-length face strips per baffle, 12 strips total. Cut three face strips per sheet and cut tip strips from wall or end offcuts.

Bill of materials

Planning quantities for Option 4 and the separate 6 in roof assembly. Confirm dimensions and stock availability before ordering.

Material or assembly	Planning quantity	Scope and fit
1/2 in exterior plywood, 4 x 8 ft	4 sheets	Two for front, one for two ends, one for ceiling and three baffle blanks. Building wall serves as backing.
2 x 3 nominal exterior framing	~90 linear ft	Front rails and five studs use about 45 ft. Balance is an allowance for returns, wall rails and blocking.
5/8 in baffle build-up allowance	1 sheet, 4 x 8 ft	Three nominal 92 7/8 x 7 7/8 in blanks if using full plywood build-up. Detail load-bearing support separately.
1 in Type 703 FRK, 24 x 48 in	24 boards / 192 ft2	About 156 ft2 gross installed before vent cutouts. Layout uses about 22 boards plus two spare.
Baffle supports and liner retention	3 support sets + 1 liner lot	Continuous cleats or brackets coordinated with wall framing. Retention and edge treatment within liner envelope.
Gaskets, sealant and fasteners	1 lot	Gasket removable front sections. Seal shell joints. Select exterior-rated fasteners and compatible sealant.
LD6 and 6 in round fittings	1 silencer + 2 elbows	Short-radius 90-degree elbows. Duct collars, joints and support hardware as required.
16 in square roof chase	1 assembly	Cover full wide silencer body. Include flashing, closure around neck and independent duct support.

Before releasing cuts: verify tree and root clearance, grade at the 12 in opening, soffit/hood dimensions and attachment locations. Check the full framed envelope against the tree. Confirm outlet flow and static pressure with the fans operating together. Nominal lined slot area is 5.71 ft2, about 1,730 ft/min at the conservative 9,900 CFM allowance.